

Capstone Worklog : Kelly

Quarter 1 summary:

I spent most of the first quarter understanding how to analysis scrnaseq data for the peripheral blood analysis and then meeting with the project leads suzana and grace to really understand what I should be looking for when doing the analysis. I used a practice dataset found online before actually doing analysis on the datasets I chose for the final analysis to really understand the theories and methodology behind looking at scRNAseq data using the scrna best practices guide online. This is due to us meeting with our project leads late from communication issues and getting mock figures done for the final presentation in the last week.

Weekly Updates:

Week 3:

- 4/13 (2 hours) - Met with grace in person ; Finding healthy pediatric peripheral blood datasets
- 4/16 (6 hours)-Finding healthy pediatric peripheral blood datasets ; received updated project goals; found 3 new healthy donor pediatric peripheral blood datasets. generated heatmaps for all these datasets;
- 4/18 (4 hours) - Worked on generating heat maps and updated slidedeck

Week 4:

- 4/22 (2 hrs) - worked on generating pediatric blood UMAPs for CD24, SIGLEC10,NECTIN4,CDH1, ST8SIA1, and B4GALNT;
- 4/23(1 hr) Meeting with Grace and Rachel

Week 5:

- 4/25 (150 mins) - Meeting with grace; and worked on making new figures for poster and got code/ graph approved by josh before putting it onto poster template; finished my section of the poster
- 4/28 Made edits on the poster before legal submittal at 4pm
- Found thymus and brain datasets for AIM 2

Week 6:

- 5/4 (240 minutes) - Worked on presentation ; meeting with grace going over presentation; Got routinely updates about the legal approval for our poster with emi and Dr. Davidson. Monthly Biology department meeting (30 minutes) with Josh, Douglas, suzana, Rachel, and grace. Finished heatmaps for AIM 1.

- 4/25 - meeting with grace; talked about AIM 2 dataset and worked on presentation
- Finished Thymus heatmaps for aim 2

Week 7:

- 5/11 (90 minutes) - Meeting with Grace ; Asked for figure explanation to be sent to all students; also asked to update project timelines to correspond with updated timelines from the updated final presentation dates.
- [180 minutes] Finished finding datasets for AIM 2 and started analysis for thymus dataset
- [240 minutes] Finished analysis of All other tissues for aim 2

Week 8:

- [60 minutes] meeting with Grace to discuss final deliverables and to check aim 2 graphs
- [240 minutes] worked on Aim 3 finding datasets and KM plots over the weekend

Week 9:

- [90 minutes] Meeting with Grace and Rachel to discuss Further analysis on Aim 3 KM plots and final presentation guidelines
- [120 minutes] worked on FRM slides draft and finished draft to submit to Rachel and Grace
- [30 minutes] finished Further KM plot analysis and sent in AIM 2&3 to get checked by josh before putting it onto slide draft

Week10:

- [60 minutes] Meeting with grace to discuss AIM 3 KM analysis changes and practice FRM slides
- [60 minutes] Meeting with grace again to practice FRM slides
- [30 minutes] Meeting with Grace and Rachel for final dress rehearsal for slides
- [45 minutes] Meeting with Josh to discuss FRM slides to better explain the methods and analysis used in preparation for the meeting & and rehearsed with Emi and Dr.Davidson at Thursday meeting.